

SHANTVEER KESTI

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SUMMARY

AI/ML Engineer with hands-on experience in PyTorch, TensorFlow, Scikit-learn, and Pandas, along with expertise in Generative AI frameworks such as LangChain, LangGraph, and Model Context Protocol (MCP). Experienced in building end-to-end AI systems, including RAG pipelines, LLM-based agents, and MCP-powered tool servers, leveraging FastAPI, Docker, and AWS for scalable deployment.

EDUCATION

ALVA'S INSTITUTE OF ENGINEERING AND TECHNOLOGY
B.E (Artificial Intelligence and Machine Learning)

DEC 2022 - MAY 2026
CGPA - 8.5

INTERNSHIPS

ASTRANOVA DYNAMICS LLP

ON SITE

ML INTERN

August 2025 - April 2026

- Worked on Machine Learning and AI-based projects in a deep-tech startup environment.
- Assisted in data preprocessing, model experimentation, and workflow implementation using Python.
- Gained hands-on exposure to Machine Learning concepts, AI tools, and development practices.
- Collaborated with team members on research-oriented technical tasks and real-world problem-solving.
- Participated in onsite project discussions, technical learning sessions, and implementation activities.

Tech : LangChain, Langgraph, LangSmith, GenAI, AgenticAI

TEACH MAVEN

REMOTE

AI/ML INTERN

March 2025 - May 2025

- Worked on AI/ML projects using Python, focusing on real-world problem-solving and data-driven solutions.
- Gained hands-on experience in machine learning concepts, model building, and data processing.
- Strengthened knowledge in statistics, data analysis, and Python programming during training and internship phases.

Tech : Python, Numpy, Pandas, ScikitLearn, Feature Engineering

PROJECTS

AI Research Agent (RAG + Long-Term Memory + Tool-Augmented LLM) - [Link](#)

- Developed a tool-enabled AI agent using LangGraph and LangChain capable of retrieving information from documents, web sources, and research papers using Retrieval-Augmented Generation (RAG).
- Implemented persistent conversational memory and streaming responses using FastAPI, PostgreSQL (PGVector), and LangSmith for observability, with a Streamlit-based interactive interface.

AI Expense Tracker using Model Context Protocol (MCP) - [Link](#)

- Developed a Local MCP Server using Python and FastMCP to enable AI assistants to manage expenses through tools such as add, list, summarize, edit, and delete, with SQLite-based storage and structured categories.
- Designed and deployed a Remote MCP Server accessible via HTTP and integrated it with a local proxy MCP architecture, allowing AI clients to securely access remote expense management tools.

RAG-Based AI Document Assistant using LangGraph - [Link](#)

- Built an end-to-end Retrieval-Augmented Generation (RAG) system using LangChain, LangGraph, FastAPI, and pgvector to enable intelligent document question answering.
- Implemented Dockerized microservices with PostgreSQL vector search and Streamlit UI, enabling scalable semantic search and contextual AI responses.

Breast Cancer Detection Using Machine Learning- [Link](#)

- Developed a machine learning model using Support Vector Machine (SVM) to classify breast tumors as benign or malignant based on 30 diagnostic features from the Breast Cancer Wisconsin dataset.
- Built an interactive Streamlit web dashboard to collect medical inputs and provide real-time predictions, integrating a Scikit-learn pipeline with a serialized Pickle model.

Data-Driven Study on the Dermatological Impact of Sunscreen Formulations - [Link](#)

- Conducted a comparative analysis of mineral and chemical sunscreen products to assess effectiveness and dermatological safety.
- Identified high-risk ingredients associated with skin irritation, providing actionable insights to support informed consumer decisions.

SKILLS

Programming: Python

Data Analysis: Numpy, Pandas, Matplotlib, Seaborn, PowerBI, Tableau, Alteryx

Machine Learning Framework: Scikit-Learn

Deep Learning Framework: Tensorflow, Keras, PyTorch

Generative AI/ LLM: RAG,Langchain, LangGraph,MCP Client, MCP Server

Vector DB: PgVector, FAISS

Databases : MySQL, SQL SERVER

Backend : FastAPI, PyDantic

Tools: Git, Docker, VS Code, Jupyter Notebook, Anaconda, Replit

Cloud: AWS(EC2 Instance)

CERTIFICATIONS

AI for Everyone - [Coursera](#)

Machine Learning Algorithms with Python - [Coursera](#)

Machine Learning II – [Coursera](#)

Data Analysis with Python - [IBM](#)

Python for Data Science - [IBM](#)

Programming, Data Structures and Algorithms - [NPTEL](#)

PUBLICATIONS

Shantveer Kesti, “A Survey on Fast Task Adaptation Techniques in Machine Learning”, IEEE Xplore, May 2025. [Link](#)

Shantveer Kesti,“SleepSensei-AI: Revolutionizing Sleep Coaching”,IEEE Xplore, February 2026. [Link](#)

OTHER ACTIVITIES

Hackathon Participant, Python Mentor (Pynocturnal Club), and Event Lead (NVIDIA TechVerse) – Explored innovative problem-solving, conducted Python sessions for juniors, and led event coordination ensuring smooth execution.